

Online Auction System Application Commands and Contracts

Note: all the `info` variables in the arguments are data transfer objects.

1 Create Auction

Command: <code>addAuction(sellerId, auctionInfo)</code>	
Preconditions	- signed-in seller (existing seller identified by <code>sellerId</code>) - no Pending Payment for seller
Postconditions	- a new Auction <code>auc</code> was created (instance creation) <code>auc</code> was initialized from <code>auctionInfo</code> (attribute modification) - the processing fee of <code>auc</code> was set (attribute modification) <code>auc</code> was added to the seller's auctions (association creation) - a new Auction Item <code>aucIt</code> was created (instance creation) <code>aucIt</code> was initialized from <code>auctionInfo</code> (attribute modification) <code>aucIt</code> was set as the item of <code>auc</code> (association creation) - if credit card information was provided in <code>auctionInfo</code> - a new Credit Card <code>ccard</code> was created (instance creation) <code>ccard</code> was initialized from the credit card information (attribute modification) <code>ccard</code> was set as the seller's credit card (association creation) <code>ccard</code> was set as the credit card of <code>auc</code> (association creation)

2 Place Bid

Command: <code>placeBid(buyerId, auctionId, bidInfo)</code>	
Preconditions	- signed-in buyer (existing buyer identified by <code>buyerId</code>) - no Pending Payment for buyer - there is an existing auction identified by <code>auctionId</code> - the auction is open
Postconditions	- a new Bid <code>abid</code> was created (instance creation) <code>abid</code> was initialized from <code>bidInfo</code> (attribute modification) <code>abid</code> was added to the buyer's bids (association creation) <code>abid</code> was added to the auction's bids (association creation)

3 Close Auction

Command: <code>closeAuction(auctionId)</code>	
Notes	• check if there were bids for the auction • submit a payment request to the Credit Service Bureau • send e-mails to the Buyer and the Seller
Preconditions	• there is an existing auction identified by <code>auctionId</code> • the auction is open • the auction deadline has passed
Postconditions	• the auction was marked as closed (attribute modification) • if the payment was not successful ▸ a pending payment notice was added to the auction's seller account (attribute modification)

4 Create Account

Command: <code>createAccount(accountInfo)</code>	
Preconditions	• <code>accountInfo</code> does not match an existing User Account
Postconditions	• a new User Account <code>acc</code> was created (instance creation) • <code>acc</code> was initialized from <code>accountInfo</code> (attribute modification) • <code>acc</code> was set as active (attribute modification) • if <code>accountInfo</code> includes <code>creditCardInfo</code> ▸ a new Credit Card <code>ccard</code> was created (instance creation) ▸ <code>ccard</code> was initialized from <code>creditCardInfo</code> (attribute modification) ▸ <code>ccard</code> was linked as the credit card of <code>acc</code> (association modification)

5 Update Account

Command: <code>updateAccount(userId, accountInfo)</code>	
Preconditions	• there is an existing user identified by <code>userId</code> • the user is active
Postconditions	• the properties of the user were modified from <code>accountInfo</code> (attribute modification) • if <code>accountInfo</code> includes <code>creditCardInfo</code> ▸ a new Credit Card <code>ccard</code> was created and initialized from <code>creditCardInfo</code> (instance creation, attribute modification) ▸ if there was a Credit Card <code>oldccard</code> linked to the user – <code>oldccard</code> was unlinked from the user (association destruction) ▸ <code>ccard</code> was set as the user's credit card (association modification) ▸ if there was a Pending Payment notice <code>pp</code> for the user and the payment was successful – <code>pp</code> was removed from the user's account (attribute modification)

6 Deactivate Account

Command: <code>deactivateAccount(accountId)</code>	
Preconditions	• there is an existing account identified by <code>accountId</code> • the account is active • the account does not have a pending payment • the account does not have a linked open auction
Postconditions	• the account was marked inactive (attribute modification)